Faculty Teaching Handbook
M.Sc. Mathematics with Artificial Intelligence & Data Science
Course Journey

Unit I → Introduction to AI → Intelligent Agents → Problem Solving → Search Techniques
Unit II → Knowledge Representation → Logic → Expert Systems
Unit III → Machine Learning → Decision Trees → Neural Networks
Unit IV → Natural Language Processing → Computer Vision
Unit V → Modern AI, Generative AI, Case Studies & Revision

Lecture 1 Introduction to Artificial Intelligence

• How many of you have used Google Maps today?
• How many watched YouTube today?
• How many used ChatGPT, Instagram or WhatsApp?
• How many unlocked your phone using Face Unlock or Fingerprint?

Most people think AI means robots. That is a myth. AI is already inside your phone, banking app, shopping app, social media, navigation system and healthcare applications. During this semester, we will not simply study AI—we will understand how intelligent systems think, learn and make decisions. We will also discover how Mathematics is the foundation of Artificial Intelligence.
By the end of this course, whenever you use ChatGPT, Google Maps or Netflix, you will understand the mathematical ideas and algorithms working behind them.

Artificial Intelligence
↓
Learning
↓
Reasoning
↓
Decision Making
↓
Problem Solving

Classroom Discussion

"Who is an intelligent person?"
INTELLIGENCE
↓
Observe
↓
Learn
↓
Understand
↓
Reason
↓
Decide
↓
Solve Problems
↓
Adapt
What is Intelligence? Intelligence is the ability to acquire knowledge, understand situations, learn from experience, reason logically, solve problems, adapt to new situations and make appropriate decisions.

Characteristics of Intelligence

- Learning
- Reasoning
- Problem Solving
- Decision Making
- Adaptation
- Creativity

What is Artificial Intelligence?

Artificial = Man-made
Intelligence = Ability to Think, Learn and Decide

Artificial Intelligence is the branch of Computer Science that develops intelligent systems capable of learning from data, reasoning, solving problems and making decisions similar to humans. OR
Artificial Intelligence is the science of making machines intelligent enough to perform tasks that normally require human intelligence.

Traditional Programming vs AI

Traditional Programming	Artificial Intelligence
Fixed instructions	Learns from data
Same output	Improves with experience
Rule-based	Data-driven
Cannot learn	Learns continuously

Traditional Programming
Input → Program → Output

Artificial Intelligence
Data → Learning → Model → Prediction → Decision

Mathematics Connection

Artificial Intelligence = Mathematics + Algorithms + Data

Core mathematical foundations include Logic, Linear Algebra, Probability, Statistics and Optimization.

Module 3 Learning Outcomes

- Recognize AI applications in everyday life.
- Explain how AI is embedded in common digital services.
- Relate AI concepts with daily activities.
- Develop curiosity through observation and discussion.

"*Imagine you wake up at 6:00 AM. Your phone unlocks using Face Unlock. Google Maps suggests the fastest route. You pay using UPI, watch YouTube, receive shopping recommendations from Amazon, ask ChatGPT a question and end the day watching Netflix recommendations. Without realizing it, you have used Artificial Intelligence dozens of times in a single day.*"
'Was AI only inside robots? Or has it already become a part of your life?'
"Where did you use AI today?"

Morning → Mobile → AI
Travel → Google Maps → AI
Shopping → Amazon → AI
Learning → ChatGPT → AI
Entertainment → Netflix/YouTube → AI

25 Real-Life Examples of AI

Application	AI Role
Google Maps	Shortest route prediction
ChatGPT	Question answering and text generation
YouTube	Video recommendation
Netflix	Movie recommendation
Amazon	Personalized shopping
Flipkart	Product recommendation
Instagram	Feed ranking
Facebook	Friend suggestions
LinkedIn	Job recommendation
Spotify	Music recommendation
Google Translate	Language translation
Gmail	Spam filtering
Face Unlock	Face recognition
Phone Camera	Scene detection
Alexa	Voice assistant
Siri	Speech recognition
Swiggy	Delivery optimization
Uber/Ola	Demand prediction
UPI Fraud Detection	Fraud detection
Banking	Credit scoring
Healthcare	Disease prediction
Agriculture	Crop monitoring
Smart Watch	Health monitoring
CCTV	Facial recognition
ISRO Satellite Images	Image analysis

AI with Mathematics:
• Google Maps → Graph Theory
• Recommendation Systems → Probability & Statistics
• ChatGPT → Linear Algebra
• Fraud Detection → Statistics
• Image Recognition → Matrix Operations
• Route Optimization → Optimization Techniques
Activity: list as many AI applications as can find on smartphones. Ask each group to explain one application and identify what makes it intelligent.

Mini Case Study

Case: One day all AI systems stop working. Discuss what will happen to banking, transport, healthcare, education, communication and social media. This activity helps students appreciate the importance of AI.

Quick Quiz

1. What is the difference between automation and AI?
2. Is Google Maps an AI application? Why?
3. Give five AI applications used every day.
4. Which AI application do you use most frequently?
5. Why is ChatGpt considered an AI system?

Homework

Prepare a one-page report titled 'Artificial Intelligence in My Daily Life'. Identify at least 20 AI applications you use in one day and explain the purpose of each in one sentence.

Module 5 Mathematics Behind Artificial Intelligence

MATHEMATICS
↓
Algorithms
↓
Data
↓
Artificial Intelligence
↓
Intelligent Applications

Why Mathematics Matters in AI

Artificial Intelligence learns patterns from data. Learning requires mathematical models, calculations and optimization. Every modern AI system—from Google Maps to ChatGPT—depends on mathematical concepts.

Core Mathematical Foundations

Mathematical Area	Role in AI	Example
Logic	Used for reasoning, inference, expert systems and decision making.	Medical expert system
Probability	Handles uncertainty, prediction, spam filtering and risk estimation.	Weather prediction
Linear Algebra	Represents data as vectors and matrices; fundamental for machine learning and deep learning.	ChatGPT embeddings
Statistics	Data analysis, model evaluation and pattern discovery.	Fraud detection
Optimization	Finds the best solution by minimizing error or maximizing performance.	Training neural networks
Graph Theory	Used in networks, search engines, recommendation systems and route planning.	Google Maps shortest path
Calculus	Optimizes learning by updating model parameters during training.	Gradient descent

Mathematics in Daily AI Applications

Application	Mathematics Used
Google Maps	Graph Theory + Optimization
Netflix	Probability + Statistics
Amazon	Statistics + Optimization
Face Recognition	Linear Algebra
ChatGPT	Linear Algebra + Probability + Optimization
Spam Detection	Probability + Statistics

Logic → Reasoning
Probability → Prediction
Linear Algebra → Data Representation
Statistics → Learning from Data
Optimization → Best Solution
Graph Theory → Intelligent Search

Classroom Activity

to identify one mathematical topic already studied and discuss where it could be useful in AI.

Research Corner

Explain that modern AI research combines Mathematics with Computer Science. Fields such as Explainable AI, Reinforcement Learning, Optimization, Cryptography and Post-Quantum Security all rely heavily on mathematical foundations.

Quick Quiz

6. Why is Linear Algebra important in AI?
7. Which branch of Mathematics helps AI deal with uncertainty?
8. Which mathematical concept is used in Google Maps?
9. Why is Optimization essential for Machine Learning?
10. True or False: AI can exist without Mathematics.

Homework

Prepare a one-page chart titled 'Mathematics Behind AI'. For each mathematical subject (Logic, Probability, Linear Algebra, Statistics, Optimization and Graph Theory), write one AI application.

Module 6 Learning Outcomes

'In the previous section we learned that Mathematics is the foundation of AI. Now let us see how AI protects our digital world. Every online payment, email, banking transaction and digital signature requires security. This is where Cryptography and Artificial Intelligence work together.'

Data
↓
Cryptography (Protection)
↓
Secure Data
↓
Artificial Intelligence (Analysis)
↓
Intelligent Decision

What is Cryptography?

Cryptography is the science of protecting information from unauthorized access. It ensures confidentiality, integrity, authentication and non-repudiation. Examples include online banking, UPI, digital signatures, HTTPS and secure messaging.

Why Combine AI and Cryptography?

Cryptography protects data, while AI extracts knowledge from data. Modern systems require both. Without cryptography, AI may receive tampered data. Without AI, cyber attacks become harder to detect.

AI in Cyber Security

Application	Role of AI
Spam Detection	Identifies malicious emails using pattern recognition
Fraud Detection	Detects unusual banking or UPI transactions
Intrusion Detection	Recognizes suspicious network activity
Malware Analysis	Classifies malicious software
Phishing Detection	Identifies fake websites and emails
User Authentication	Detects abnormal login behaviour

AI and Digital Signatures

Digital signatures verify the authenticity of digital documents. AI does not replace digital signatures; instead, it complements them by detecting forged signatures, identifying unusual signing behaviour, and helping classify fraudulent documents.

AI vs Cryptography

Aspect	Artificial Intelligence	Cryptography
Primary Goal	Learn and make decisions	Protect information
Input	Data	Plaintext / Keys
Output	Prediction / Decision	Encrypted or authenticated data
Focus	Intelligence	Security
Nature	Adaptive	Mathematical security algorithms

Post-Quantum Cryptography (PQC)

Quantum computers threaten many classical public-key cryptographic systems. Post-Quantum Cryptography develops algorithms that remain secure against quantum attacks. Current NIST-standardized approaches include lattice-based cryptography. AI can assist researchers by optimizing parameter selection, analyzing attacks and discovering implementation anomalies.

Research Corner : current research topics:
• AI for Intrusion Detection Systems
• AI-assisted Malware Classification
• AI in Fraud Analytics
• Privacy-Preserving Machine Learning
• Federated Learning
• AI for Post-Quantum Cryptography
• Explainable AI for Security
Cryptography → Protects Data
Artificial Intelligence → Learns from Data
Together → Secure Intelligent Systems

Classroom Activity
to identify five places where both AI and Cryptography are used together (e.g., UPI, online banking, Aadhaar authentication, e-commerce, secure email). Discuss why both technologies are necessary.

Quick Quiz
11. What is the main purpose of Cryptography?
12. Can AI replace Cryptography? Why or why not?
13. Give two examples of AI in Cyber Security.
14. What is a Digital Signature?
15. What is Post-Quantum Cryptography?

Homework
Prepare a two-page report on 'Applications of Artificial Intelligence in Cyber Security'. Include one recent example related to fraud detection, malware analysis or digital payments.

Teacher Tips
- Clarify that AI and Cryptography complement each other rather than compete.
- If students ask about ChatGPT security, explain that encryption protects communication while AI processes information.
- Use UPI and online banking as familiar examples.
- Briefly introduce PQC without going into mathematical details in the first lecture.

Module 8
Section A: Multiple Choice Questions

Q.No.	Question	A/B/C/D (Correct option in text below)	Answer	Bloom
1	Artificial Intelligence is primarily concerned with	A:Making machines perform intelligent tasks B:Operating systems C:Computer networking D:Web design	A	Remember
2	Which of the following is an example of Narrow AI?	A:AGI B:Google Maps C:Human Brain D:ASI	B	Remember
3	Which branch of Mathematics is fundamental for route optimization?	A:Graph Theory B:Chemistry C:Biology D:History	A	Remember

4	ChatGPT is mainly an example of	A:Spreadsheet B:Large Language Model C:Compiler D:Database	B	Remember
5	AI systems primarily learn from	A:Data B:Luck C:Random guessing D:Electricity	A	Remember
6	Which one is NOT a characteristic of human intelligence?	A:Reasoning B:Creativity C:Unlimited memory D:Adaptability	C	Remember
7	Which technology protects information?	A:Cryptography B:Compiler C:HTML D:CSS	A	Remember
8	Face Unlock mainly uses	A:Computer Vision B:Database C:Sorting D:Compression	A	Remember
9	Current AI systems are mostly	A:ANI B:AGI C:ASI D:None	A	Remember
10	Optimization in AI is used to	A:Improve model performance B:Print documents C:Store files D:Increase monitor size	A	Remember

Section B: Short Answer Questions (2–5 Marks)

16. Define Intelligence.
17. Define Artificial Intelligence.
18. List any five applications of AI.
19. Differentiate Traditional Programming and AI.
20. What is Weak AI?
21. What is AGI?
22. Why is Mathematics important in AI?
23. What is Cryptography?
24. Write any four characteristics of intelligence.
25. Explain AI in one real-life application.

Section C: Long Answer Questions (10–15 Marks)

26. Explain the concept of Artificial Intelligence with suitable examples.
27. Compare Human Intelligence and Artificial Intelligence.
28. Discuss the role of Mathematics in Artificial Intelligence.
29. Explain AI applications in healthcare, education and cyber security.
30. Describe ANI, AGI and ASI with suitable examples.

Section D: Viva Questions

- What inspired the development of AI?
- Can AI think exactly like humans?
- Why is ChatGPT considered AI?

- Is Calculator an AI system? Why?
- What is the difference between learning and memorization?
- How does Google Maps use AI?
- What is the relationship between AI and Cryptography?
- What is Post-Quantum Cryptography?

Expected Viva Discussion Points

Encourage students to answer with examples rather than definitions. Ask follow-up questions such as 'Can you give a real-life example?' or 'Which branch of Mathematics supports this application?'

Lecture 1 – Part 9
Assignments, Classroom Discussion, Group Activities & Homework

Module 9

In-Class Activity 1: AI Around Me

Duration: 5 minutes
to write five AI applications they used from morning until they reached the university. Invite students to explain why each example is actually AI.

In-Class Activity 2: Group Brainstorming

Divide the class into five groups.
Assign one sector to each group:
1. Healthcare
2. Education
3. Agriculture
4. Banking
5. Transportation

Each group identifies at least five AI applications and presents them in two minutes.

In-Class Activity 3: Think-Pair-Share

Question: 'Can Artificial Intelligence ever replace human intelligence completely?'
Students think individually (2 min), discuss with a partner (3 min), then share conclusions.

Mini Case Study

Case: A hospital introduces an AI system to detect diseases from X-ray images.

Discussion Questions:
• What are the advantages?
• What are the risks?
• Should the final decision always be taken by a doctor?
Conclude that AI supports experts but human judgment remains essential.

Assignment 1 (Individual)
Title: Artificial Intelligence in My Daily Life

Write a 2-page report identifying at least 20 AI applications used in one day. For each application mention:
• Where it is used
• What problem it solves
• What type of intelligence it demonstrates

Assignment 2 (Research Based)
Prepare a presentation (5–7 slides) on one AI application in any domain such as Healthcare, Finance, Agriculture, Education, Defence or Space Technology.

Assignment 3 (Mathematics Connection)
Prepare a table showing how Logic, Probability, Linear Algebra, Statistics, Optimization and Graph Theory contribute to Artificial Intelligence.

Homework
31. Define Intelligence in your own words.
32. Define Artificial Intelligence.
33. List 25 real-life AI applications.
34. Compare Traditional Programming and AI.
35. Write the difference between ANI, AGI and ASI.
36. Explain why Mathematics is important in AI.
37. Identify one AI application used in Cyber Security.

Reflective Learning Questions
- What surprised you most in today's lecture?
- Which AI application do you use most often?
- Which topic would you like to explore further?
- How do you think AI will change your future career?

Extension Activity
to install or explore one AI tool (e.g., ChatGPT, Google Lens, Google Translate, Microsoft Copilot or Gemini) and prepare a one-page reflection on how it uses AI.

Lecture 1 – Part 10

Teacher Tips, Common Mistakes, Lecture Summary & Bridge to Lecture 2Common Student Misconceptions

Misconception	Clarification
AI means robots.	AI is present in software, apps and services.
ChatGPT is AGI.	ChatGPT is an advanced Narrow AI system.
AI can replace all humans.	AI augments humans; many tasks still require human judgment.
Programming alone is enough for AI.	Strong mathematical foundations are essential.

One-Page Lecture Summary

- Intelligence = Learn + Reason + Decide + Adapt
- Artificial Intelligence = Machines performing tasks requiring human intelligence
- Current AI = Mostly Narrow AI (ANI)
- Mathematics is the foundation of AI
- AI + Cryptography = Secure Intelligent Systems
- AI is already part of everyday life

PPT Slide Mapping

Slides	Content
1-2	Title & Motivation
3-5	AI Around Us
6-8	Intelligence
9-11	Artificial Intelligence
12-14	Human vs AI
15-17	Mathematics Behind AI
18-19	AI & Cryptography
20	Summary & Homework

Bridge to Lecture 2

'Today we understood what Artificial Intelligence is. But how did AI begin? Who introduced the term Artificial Intelligence? What is the Turing Test? Who was Alan Turing? Why did AI experience periods known as AI Winters? In the next lecture we will explore the fascinating history and evolution of Artificial Intelligence.'